

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Winter Examination -2023**

Course: B. Tech. Branch : Electronics Engineering and Allied Semester : VI
Subject Code & Name: BTETC601_Y22 Antennas and Wave Propagation

Max Marks: 60**Date: 16/01/2024****Duration: 3 Hr.****Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Describe the layers of the Earth's atmosphere and their impact on radio wave propagation.	Understand	6
B) Define MUF, A high frequency radio link has to be established between two points at a distance of 2500 km on earth's surface. Consider ionospheric height to be 220 km and its critical frequency 4 MHz, Calculate the MUF for given path.	Evaluate	6
C) Derive an expression for Friis Transmission Equation, and explain its significance on signal propagation in wireless communication systems?	Analyze	6
Q.2 Solve Any Two of the following.		12
A) Explore the various types of antennas and their applications in different communication scenarios.	Analyze	6
B) Define and elaborate on terms such as radiation pattern, radiation power density, radiation intensity, directivity, and gain.	Remember	6
C) Derive an expression for the fields radiated by infinitesimal dipole antenna.	Analyze	6
Q. 3 Solve Any Two of the following.		12
A) Provide a detailed explanation of a log periodic antenna along with its unique features, Also explain its different applications.	Understand	6
B) Compare the characteristics of a binomial and Dolph Tchebyshev antenna array. Explain the design principles behind each type and how they affect the antenna's performance.	Analyze	6
C) Discuss the principle of pattern multiplication in an N-element linear array. How does the radiation pattern evolve as the number of elements increases?	Understand	6
Q.4 Solve Any Two of the following.		12
A) Explain the advantages of smart antennas in wireless communication systems.	Analyze	6
B) Compare adaptive and fixed beamforming techniques along with the distinguishing features of each, and in what scenarios are they most effective?	Analyze	6

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| C) Explain different challenges associated with the implementation of smart antennas in communication systems. | Analyze | 6 |
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Q. 5 Solve Any Two of the following. **12**

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| A) Write a short note on (a) Rhombic Antenna and (b) V antenna. | Remember | 6 |
| B) Provide a detailed explanation of the microstrip antenna's structure, and discuss the different methods used for feeding this type of antenna. | Understand | 6 |
| C) Explain the structural characteristics and dimensions of antennas with parabolic reflectors. Discuss the features and applications that make antennas with parabolic reflectors effective for long-range communication. | Analyze | 6 |

***** End *****